

Week 2: Statistical Properties of Financial Data

Financial Assets

- ▶ Financial assets do not have physical worth (they are mostly pieces of paper). They derive their financial value from the fact that they represent a contractual claim on a stream of cash flows.
- ▶ Equities/stocks: give the owner an equity stake in a company (a share) and a claim on company's assets and earnings. These claims can be bought and sold on stock markets.
- ▶ Stocks give the owner the right to a payment which represents the distribution of some of the company's earnings, which is known as a dividend.
- ▶ The dividend is usually expressed as the amount each share receives or as a percentage of the current market price, referred to as dividend yield.

Stock Certificate

Figure Sapir Consolidated Airlines Stock



Equity Prices

- ▶ The price of an equity security is defined as the amount at which a transaction can occur (quoted price) or has occurred (historical price).
- ▶ Recording a “price” for the purposes of doing econometric analysis is not entirely straightforward. Multiple ways to define price:
 1. day’s closing price
 2. day’s opening price
 3. some markets even have overnight prices
- ▶ The adjusted closing price is generally used as the price because it accounts for stock splits and dividends. Therefore, we will generally use the adjusted closing price.

What does price tell us?

1. The goal of investing is to earn a profit. Price doesn't equal profit.
2. The most meaningful measure of profit for investors is return.
3. Return measures the appreciation of an asset.
4. It is measured as some difference between the current value of the asset, i.e., the current price, and its initial value, i.e., the price you paid for the asset.

Returns

- ▶ A financial return is a measure of the profit from investing in a financial asset.
- ▶ It accounts for the capital gain or loss due to the price change over the holding period of the asset.
- ▶ Also, it is impacted by the stream of cash flows (dividends) that occur over the course of the holding period.
- ▶ Data on prices are usually recorded at daily, weekly or monthly frequency, so holding period of the investment is limited to a multiple of this frequency.
- ▶ An investor could theoretically hold a financial asset forever, assuming the company never goes bankrupt.

Returns

Let P_t be the price of an asset at time t .

1. The simple **net** return over the period from $t - 1$ to t is

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}}.$$

2. The simple **gross** return is

$$\frac{P_t}{P_{t-1}} = 1 + R_t$$

Consider a one-month investment in Microsoft stock. Suppose you buy the stock in month $t - 1$ (e.g., end of February) at $P_{t-1} = \$85$ and sell the stock the next month (e.g., end of March) for $P_t = \$90$.

The one-month simple net and gross returns are then:

$$R_t = \frac{\$90 - \$85}{\$85} = \frac{\$90}{\$85} - 1 = 1.0588 - 1 = 0.0588$$

$$1 + R_t = 1.0588$$

The one-month investment in Microsoft yielded a 5.88% per month return.

Alternatively, \$1 invested in Microsoft stock in month $t - 1$ grew to \$1.0588 in month t .

Multi-period returns

The simple two-month return on an investment in an asset between months $t - 2$ and t is defined as:

$$R_t(2) = \frac{P_t - P_{t-2}}{P_{t-2}} = \frac{P_t}{P_{t-2}} - 1.$$

Writing

$$\frac{P_t}{P_{t-2}} = \frac{P_t}{P_{t-1}} \frac{P_{t-1}}{P_{t-2}}$$

the two-month return can be expressed as:

$$\begin{aligned} R_t(2) &= \frac{P_t}{P_{t-1}} \frac{P_{t-1}}{P_{t-2}} - 1 \\ &= (1 + R_t)(1 + R_{t-1}) - 1. \end{aligned}$$

Multi-period returns

Then the simple two-month **gross** return becomes:

$$1 + R_t(2) = (1 + R_t)(1 + R_{t-1}) = 1 + R_{t-1} + R_t + R_{t-1}R_t$$

Hence, the two-month **net** return is

$$R_t(2) = R_{t-1} + R_t + R_{t-1}R_t$$

If, however, R_{t-1} and R_t are close to zero then $R_{t-1}R_t \approx 0$ and

$1 + R_t(2) \approx 1 + R_{t-1} + R_t$ so that

$$R_t(2) = R_{t-1} + R_t.$$

Multi-period returns

Continuing with the previous example, suppose that the price of Microsoft stock in month $t - 2$ is \$80.

The two-month net return is:

$$R_t(2) = \frac{\$90 - \$80}{\$80} = \frac{\$90}{\$80} - 1 = 1.1250 - 1 = 0.1250$$

or 12.50% per two-month.

Multi-period returns

The two one-month returns are:

$$R_{t-1} = \frac{\$85 - \$80}{\$80} = 1.0625 - 1 = 0.0625$$

$$R_t = \frac{\$90 - \$85}{\$85} = 1.0588 - 1 = 0.0588$$

The two-month gross return

$$1 + R_t(2) = (1.0588)(1.0625) = 1.1250$$

Returns

In general, return of holding an asset for k periods, $R_t(k)$, is then

$$\begin{aligned} R_t(k) &= \frac{P_t}{P_{t-k}} - 1 \\ &= \frac{P_t}{P_{t-1}} \times \frac{P_{t-1}}{P_{t-2}} \times \cdots \times \frac{P_{t-k+2}}{P_{t-k+1}} \times \frac{P_{t-k+1}}{P_{t-k}} - 1 \\ &= (1 + R_t) \times (1 + R_{t-1}) \times \cdots \times (1 + R_{t-k+2}) \times (1 + R_{t-k+1}) - 1 \\ &= \prod_{j=0}^{k-1} (1 + R_{t-j}) - 1 \end{aligned}$$

Log returns

The **log return** is denoted by r_t and is defined as

$$r_t = \log(1 + R_t) = \log\left(\frac{P_t}{P_{t-1}}\right) = \log(P_t) - \log(P_{t-1}).$$

Log returns are approximately equal to net returns because if x is small, then $\log(1 + x) \approx x$.

Multi-period returns

Consider the two-month log return defined as:

$$\begin{aligned}r_t(2) &= \log(P_t) - \log(P_{t-2}) \\&= [\log(P_t) - \log(P_{t-1})] + [\log(P_{t-1}) - \log(P_{t-2})] \\&= r_t + r_{t-1}\end{aligned}$$

Hence the two-month log return is just the sum of the two one-month log returns.

Multi-period returns

The k -period log return is

$$\begin{aligned}r_t(k) &= \log P_t - \log P_{t-k} \\&= (\log P_t - \log P_{t-1}) + (\log P_{t-1} - \log P_{t-2}) + \cdots \\&\quad + (\log P_{t-k+1} - \log P_{t-k}) \\&= r_t + r_{t-1} + \cdots + r_{t-(k-1)} \\&= \sum_{j=0}^{k-1} r_{t-j}\end{aligned}$$

The k -period log return is simply the sum of the single period log returns over the pertinent period.

Compute two-month log returns

Using the price and return data from the previous example, the log monthly return on Microsoft stock can be computed in two ways:

$$r_t = \ln(1.0588) = 0.0571$$

or

$$r_t = \ln(90) - \ln(85) = 4.4998 - 4.4427 = 0.0571$$

Compute two-month log returns

Using the data from the previous example, the two-month log return on Microsoft stock can be computed in two equivalent ways.

The first way uses the difference in the logs of P_t and P_{t-2}

$$r_t(2) = \ln(90) - \ln(80) = 4.4998 - 4.3820 = 0.1178.$$

Compute two-month log returns

The second way uses the sum of the two one-month log returns. Here

$$r_t = \ln(90) - \ln(85) = 0.0571$$

$$r_{t-1} = \ln(85) - \ln(80) = 0.0607$$

so that

$$r_t(2) = 0.0571 + 0.0607 = 0.1178$$

Dividends

What about dividends?

In the Microsoft example, we assume that no dividend is paid between months $t - 1$ and t .

For multi-period returns, we again assume that no dividend is paid between months $t - 2$ and t .

Dividends

Adjusting the calculation of returns for the payment of dividends, denoted by D_t , between time $t - 1$ and t is relatively straightforward.

For simple net and gross returns

$$R_t = \frac{P_t + D_t - P_{t-1}}{P_{t-1}} = \frac{P_t}{P_{t-1}} + \frac{D_t}{P_{t-1}} - 1$$

$$1 + R_t = \frac{P_t + D_t}{P_{t-1}} = \frac{P_t}{P_{t-1}} + \frac{D_t}{P_{t-1}}$$

Relative price (P_t/P_{t-1}) and the dividend yield (D_t/P_{t-1}).

Dividends

- ▶ For log returns

$$r_t = \log(1 + R_t) = \log\left(\frac{P_t + D_t}{P_{t-1}}\right) = \log\left(\frac{P_t}{P_{t-1}} + \frac{D_t}{P_{t-1}}\right)$$

- ▶ Much of the unit will ignore the issue of dividends.
- ▶ This stems from the reality that dividends are paid relatively infrequently.
- ▶ Also they constitute a minor proportion of the overall return compared with price movements.

Portfolio Returns

Often concerned not with the return to a single asset but rather the return to a **portfolio** of financial assets.

Consider a portfolio with only two assets whose portfolio shares are w_1 and w_2 respectively, with the normalization

$$w_1 + w_2 = 1.$$

Using simple gross returns for each asset, the value of the portfolio between $t - 1$ and t may be calculated as

$$\begin{aligned}P_t &= w_1 P_{t-1} (1 + R_{1t}) + w_2 P_{t-1} (1 + R_{2t}) \\&= P_{t-1} (w_1 (1 + R_{1t}) + w_2 (1 + R_{2t}))\end{aligned}$$

We can rearrange as

$$(1 + R_{pt}) \equiv \frac{P_t}{P_{t-1}} = w_1 (1 + R_{1t}) + w_2 (1 + R_{2t})$$

Expanding yields

$$1 + R_{pt} = w_1 + w_2 + w_1 R_{1t} + w_2 R_{2t}$$

so that

$$R_{pt} = w_1 R_{1t} + w_2 R_{2t}.$$

Portfolio Returns

In simple returns, the portfolio return is just the weighted sum of individual returns.

$$R_{pt} = w_1 R_{1t} + w_2 R_{2t}.$$

This is true for N returns

$$R_{pt} = \sum_{i=1}^N w_i R_{it},$$

with

$$\sum_{i=1}^N w_i = 1.$$

Portfolio Returns: Log Returns

This result does not extend to the case of log returns

$$r_{pt} = \log(1 + R_{pt}) = \log\left(1 + \sum_{i=1}^N w_i R_{it}\right) \neq \sum_{i=1}^N w_i r_{it}.$$

How to combine log returns into the portfolio return?

Portfolio Returns: Log Returns

Recall

$$r_t = \log(1 + R_t) = \log\left(\frac{P_t}{P_{t-1}}\right).$$

Take the exponential of both sides to give:

$$e^{r_t} = 1 + R_t = \frac{P_t}{P_{t-1}}$$

Rearranging we get

$$P_t = P_{t-1}e^{r_t}$$

Portfolio Returns: Log Returns

The value of the portfolio between $t - 1$ and t may be calculated as

$$P_t = w_1 P_{t-1} e^{r_{1t}} + w_2 P_{t-1} e^{r_{2t}}$$

so that

$$r_{pt} \equiv \log \left(\frac{P_t}{P_{t-1}} \right) = \log (w_1 e^{r_{1t}} + w_2 e^{r_{2t}})$$

For N assets,

$$r_{pt} = \log \left(\sum_{i=1}^N w_i e^{r_{it}} \right)$$

Portfolio Returns: Log Returns

- ▶ Most of the time we **simply ignore** this formula

$$r_{pt} = \log \left(\sum_{i=1}^N w_i e^{r_{it}} \right)$$

- ▶ Instead we define

$$r_{pt} := \sum_{i=1}^N w_i r_{it}.$$

- ▶ We simply take a weighted aggregate of these log returns to find portfolio returns.

Stock Indices

- ▶ An aggregate summary measure of the performance of the stock market as a whole is known as a stock market index.
- ▶ Indices combine a selection of (a large number of) stocks in a particular way to create a portfolio.
- ▶ Stock market indices are price indices. The computation of returns to the index can be performed in exactly the same way as if it were a single stock.

Stock Indices: Two common ways of construction:

1. Price-weighted index (PWI) is a stock market index in which each stock is given a weight based on its price per share. This means that stocks with a higher share price have a larger effect on the index.
2. Value-weighted index (VWI) is a stock market index in which each stock is given a weight based on its total market value, usually measured by market capitalisation.

Stock Indices

- ▶ Deutscher Aktien Index (DAX): comprises the 30 largest German companies that trade on the Frankfurt Stock Exchange (VWI).
- ▶ Dow Jones Industrial Average (DJ or DJIA) Index: computed using 30 prominent United State corporations (PWI).
- ▶ Financial Times Stock Exchange 100 (FTSE) Index: 100 largest companies listed on the London Stock Exchange (VWI).
- ▶ Hang Seng (HSX) Index: 40 of the largest companies that trade on the Hong Kong Exchange (VWI).

Stock Indices

- ▶ Nikkei 225 (SKX) Index: 225 prominent companies listed on the Tokyo Stock Exchange (PWI).
- ▶ Standard and Poors Composite 500 (S&P500 or S&P) index: 500 large companies having common stock listed on the NYSE, NASDAQ, or the Cboe BZX Exchange. It is computed by summing the market value of the outstanding equity in each firm in the index (VWI).

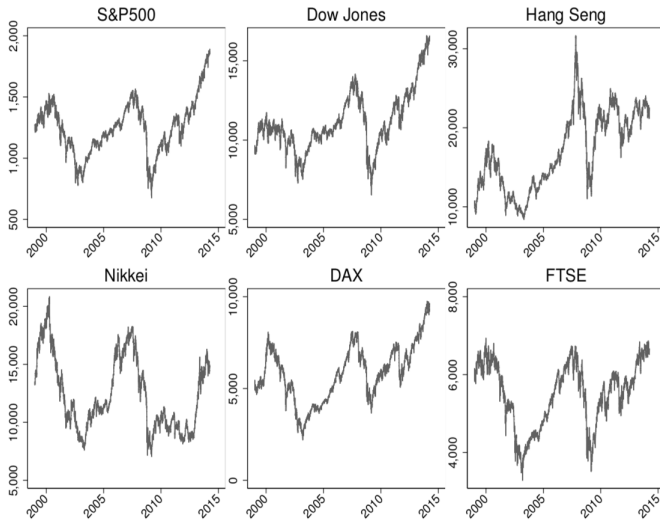


Figure 1.2: Daily observations on six international stock market indices for the period 4 January 1999 to 2 April 2014.

What Do Investors Care About

Intuitively, there are at least four factors an investor would like to know when considering investing in a risky asset.

1. The expected return from investing in the asset.
2. The uncertainty surrounding the value of, or payoff from, the investment in the asset. In other words, the chance that the actual return on an investment may be very different than the expected return.
3. Whether or not the extreme returns are above or below the expected value.
4. The relative likelihood of occurrence of extreme returns (positive or negative).

What tools do we need?

To explain these specific behaviors, we must first properly define them. We will revisit some concepts of probability and statistics.

Recall:

- ▶ the different types of random variables: continuous and discrete
- ▶ the “moments”, or expectations, of different random variables, and how they are estimated from sample data

Random Variables

Random Variables are numbers that we are not sure about but we might have some idea of how to describe its potential outcomes.

Consider the price of Microsoft stock next month, P_{t+1} . Since P_{t+1} is not known with certainty today (t), we can consider it as a random variable.

The two most common classes of random variables are called:

1. Discrete
2. Continuous

Discrete Random Variables

Definition

A discrete random variable X is one that can take on a finite number of n different values $S_X = x_1, x_2, \dots, x_n$.

Definition

The probability mass function (**pmf**) of a discrete random variable, denoted $p(x)$, is a function such that

$$p(x) = \Pr(X = x)$$

.

Bernoulli

A Bernoulli random variable takes values $X = 1$ with probability p and $X = 0$ with probability $1 - p$.

Bernoulli random variables are often used to model “success” or “failure”, where success is loosely defined.

For example, a success could be a stock price going up, a coin landing on heads, a bull market or a corporate default and a failure could be a stock price going down, a coin landing on tails, a bear market or a corporate solvency.

Often, denoted as $X \sim \text{Bern}(p)$.

Example

Let r_t be the log return on the S&P500 index at time t .

Let $X = 1$ if $r_t > 0$ and $X = 0$ else.

$X \sim \text{Bern}(p)$

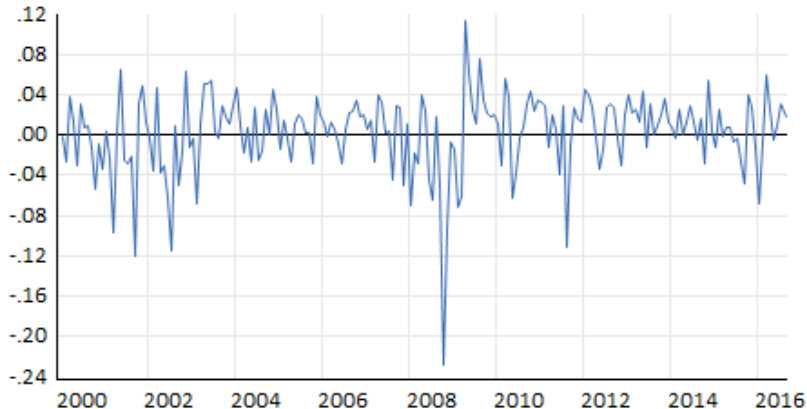
How would you estimate p ?

Mathematically, for n denoting the total number of observations, we can estimate p by

$$\hat{p} = \frac{\#\{r_t > 0\}}{n}$$

$\#\{r_t > 0\}$ is the count of positive returns.

S&P 500 monthly log returns, January 2000 to August 2016



We have $n = 200$, $\#\{r_t > 0\} = 121$ and hence $\hat{p} = 0.605$

Continuous Random Variables

Definition

A continuous random variable X is one that can take on any real value. That is, $S_X = x : x \in \mathbb{R}$

Definition

The probability density function (**pdf**) of a continuous random variable X is a nonnegative function f , defined on the real line, such that for any interval A :

$$\Pr (X \in A) = \int_A f(x) dx$$

That is, $\Pr (X \in A)$ is the area under the probability curve over the interval A .

Continuous Random Variables: Normal

The most famous continuous random variable is the “normal random variable”. Often, denoted as $X \sim N(\mu, \sigma^2)$.

A normal random variable X has pdf

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}.$$

μ is the location parameter; σ is the scale parameter.

The standard normal distribution

Special case is the “standard normal”, $\mu = 0$ and $\sigma = 1$.

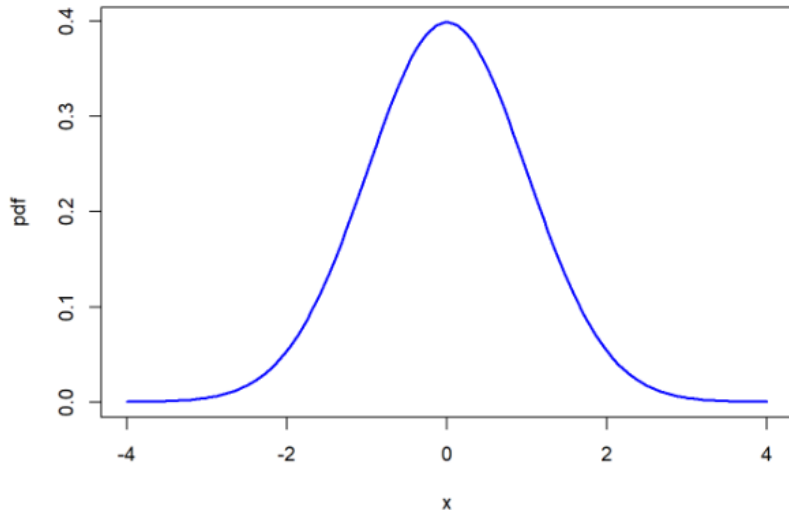
If a random variable X follows a standard normal distribution then we often write $X \sim N(0, 1)$ as short-hand notation.

The pdf of a standard normal random variable is given by:

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$$

with $-\infty \leq x \leq \infty$.

Standard normal density



Normal distribution

The normal distribution has the annoying feature that the area under the normal curve cannot be evaluated analytically.

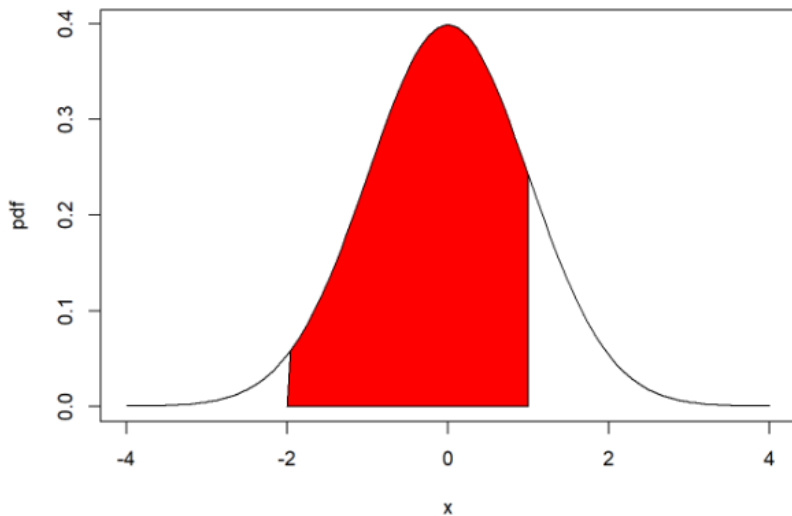
That is, $\Pr(X \in A)$ is the area under the probability curve over the interval A :

$$\Pr(a \leq X \leq b) = \int_a^b \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} dx$$

does not have a closed form solution.

Areas under the normal curve, in one form or another, are given in tables in every introductory statistics book and standard statistical software can be used to find these areas.

$\Pr(-2 \leq X \leq 1)$ is given by the area under the probability curve.



Moments / Expectations

Expectations

- ▶ We'd often like to summarize the most important features of a random variable.
- ▶ In financial econometrics, this will often be mean, variance, skewness and kurtosis.
- ▶ To understand these quantities, we must first talk about expectations.
- ▶ An expectation is the average of a random variable and summarises the central tendency of a random variable.

Expected Value: Definitions

Discrete Random Variable

If X is a discrete random variable with **pmf** $p(x)$, then the expected value of X , denoted $\mathbb{E}[X]$, is given by

$$\mu = \mathbb{E}[X] = \sum_{x \in S_X} x \cdot p(x).$$

Continuous Random Variable

If X is a continuous random variable with **pdf** $f(x)$, then the expected value of X , denoted $\mathbb{E}[X]$, is given by

$$\mu = \mathbb{E}[X] = \int_{-\infty}^{\infty} x \cdot f(x) dx.$$

Variance and Standard Deviation

For a discrete or continuous random variable X , the variance and standard deviation of X are defined as:

$$\sigma^2 = \text{var}(X) = \mathbb{E}[(X - \mu)^2]$$

$$\sigma = \text{sd}(X) = \sqrt{\sigma^2} = \sigma.$$

Because σ^2 represents an average squared deviation, it is not in the same units as X .

The standard deviation of X is the square root of the variance and is in the same units as X .

Example

Let $X \sim N(0, 1)$, then $\mathbb{E}[X] = 0$ and $\mathbb{V}[X] = 1$

Example

Let $X \sim N(\mu, \sigma^2)$, then $\mathbb{E}[X] = \mu$ and $\mathbb{V}[X] = \sigma^2$

Example

Let $X \sim \text{Bern}(p)$, then $\mathbb{E}[X] = p$ and $\mathbb{V}[X] = p(1 - p)$

Standardizing a random variable

Let X be a normal random variable with $E(X) = \mu$ and $\text{var}(X) = \sigma^2$.
That is

$$X \sim N(\mu, \sigma^2).$$

Define a new random variable Z as

$$Z = \frac{X - \mu}{\sigma} \sim N(0, 1).$$

This transformation is called standardizing the random variable X (see next slide).

Standardizing a random variable

Re-write

$$Z = \frac{X - \mu}{\sigma} = \frac{1}{\sigma}X - \frac{\mu}{\sigma}$$

and so

$$E(Z) = \frac{1}{\sigma}E(X) - \frac{\mu}{\sigma} = \frac{1}{\sigma}\mu - \frac{\mu}{\sigma} = 0.$$

$$\text{var}(Z) = \frac{1}{\sigma^2}\text{var}(X) = \frac{1}{\sigma^2}\sigma^2 = 1.$$

In short, if $X \sim N(\mu, \sigma^2)$ then $Z \sim N(0, 1)$.

Computing probabilities using standard normal

Let $X \sim N(2, 4)$ and suppose we want to find $\Pr(X > 5)$ but we only know probabilities associated with a standard normal random variable $Z \sim (0, 1)$.

We solve the problem by standardizing X as follows:

$$\Pr(X > 5) = \Pr\left(\frac{X - 2}{\sqrt{4}} > \frac{5 - 2}{\sqrt{4}}\right) = \Pr\left(Z > \frac{3}{2}\right) = 0.0668$$

Moments

Recall the four features of an asset return an investor would like to know:

1. The expected return from investing in an asset.
2. The risk associated with an investment.
3. The likelihood of extreme returns being above or below the mean return.
4. The likelihood of extreme returns.

Each of the above can be related to some “expectation or moment” of returns.

Moments

For each integer k , the **k -th moment** of X is

$$\mu_k = \mathbb{E}[X^k].$$

The **k -th central moment** of X is

$$\bar{\mu}_k = \mathbb{E}[(X - \mu_1)^k].$$

The **k -th standardized moment** of X is given by

$$\bar{\mu}_k^s = \mathbb{E} \left[\left(\frac{X - \mu_1}{\sqrt{\bar{\mu}_2}} \right)^k \right].$$

Examples

Example

Let $X \sim N(0, 1)$, then $\mathbb{E}[X] = 0$ and $\mathbb{E}[X^2] = 1$

Example

Let $X \sim N(\mu, \sigma^2)$, then $\mathbb{E}[X] = \mu$ and $\mathbb{E}[X^2] = \sigma^2 + \mu^2$

Example

Let $X \sim \text{Bern}(p)$, then $\mathbb{E}[X] = p$ and $\mathbb{E}[X^2] = p(1 - p) + p^2 = p$

How to estimate these Moments?

- ▶ Moments of asset returns are of critical importance for financial econometrics.
- ▶ The “expected return” can be measured by the mean: $\mathbb{E}[r_t]$.
- ▶ The “risk associated with an investment” can be measured by the variance: $\bar{\mu}_2 = \mathbb{E}[(r_t - \mu_1)^2]$.

Skewness

Likelihood of extremes above or below mean: referred to as **skewness**.

Measures whether more extreme positive or negative returns are likely to occur.

Skewness is more commonly measured by

$$\bar{\mu}_3^s = \mathbb{E} \left[\left(\frac{r_t - \mu_1}{\sqrt{\bar{\mu}_2}} \right)^3 \right].$$

1. If $\bar{\mu}_3^s < 0$, the distribution is negatively skewed.
2. For stock returns data, **negative skewness** is more likely than **positive skewness**, and hence large negative returns more likely than large positive returns.

Kurtosis

- ▶ Likelihood of extremes: referred to as the **kurtosis**.
- ▶ This measures the likelihood of extreme events, either positive or negative.
- ▶ Extreme events can be thought of as points far away from the center of the distribution, i.e., the mean.
- ▶ Kurtosis is measured by the fourth standard moment

$$\bar{\mu}_4^s = \mathbb{E} \left[\left(\frac{r_t - \mu_1}{\sqrt{\bar{\mu}_2}} \right)^4 \right].$$

- ▶ However, by itself it is not enough. Needs a comparison.

Kurtosis

Since kurtosis deals with the mass away from the center of a distribution, then a “large” value of kurtosis should imply that the random variable has “tails” that are relatively thick.

Compare this number relative to a standard normal distribution.

Kurtosis of the standard normal random variable is exactly three.

Often represent the kurtosis through “excess kurtosis”:

$$\text{Ex. Kurt} = \bar{\mu}_4^s - 3.$$

1. Ex. Kurt < 0 , thinner tails than normal.
2. Ex. Kurt > 0 , thicker tails than normal.

Moment Estimators

- ▶ Clearly, we don't know the distribution of asset returns, or its population moments.
- ▶ However, we can estimate the “moments” of this distribution using sample data.
- ▶ There are common summary statistics that can be used to estimate $\bar{\mu}_1, \bar{\mu}_2, \bar{\mu}_3^s, \bar{\mu}_4^s$.

Expected Return

- ▶ An estimator of an expectation is its sample average.
- ▶ So, how would we estimate expected return?
- ▶ Sample mean:

$$\bar{r} := \frac{1}{T} \sum_{t=1}^T r_t.$$

- ▶ It is a good estimator of location.
- ▶ However, it is inappropriate if data are trending / non-stationary.

Uncertainty?

- ▶ Uncertainty is related to the potential fluctuations of an asset.
- ▶ So, how would we normally measure uncertainty?
- ▶ Sample variance:

$$s^2 := \frac{1}{T-1} \sum_{t=1}^T (r_t - \bar{r})^2 .$$

- ▶ Estimator of uncertainty.

Uncertainty?

- ▶ Sample standard deviation (std dev)

$$s := \sqrt{\frac{1}{T-1} \sum_{t=1}^T (r_t - \bar{r})^2}.$$

- ▶ It is often used as a measure of the riskiness of an investment.
- ▶ Often called the volatility of a financial returns.
- ▶ The standard deviation has the same scale as returns (the variance has the scale of returns squared) and is therefore easily interpretable.

Positive or Negative Extreme Returns?

- ▶ Need a measure of whether there are more **large** negative or positive returns.
- ▶ So, how would we normally measure this asymmetry?
- ▶ Sample Skewness

$$\text{SK} = \frac{1}{T} \sum_{t=1}^T \left(\frac{r_t - \bar{r}}{s} \right)^3 .$$

- ▶ Here, s stand for standard deviation where:

$$s^2 := \frac{1}{T-1} \sum_{t=1}^T (r_t - \bar{r})^2 .$$

Sample Skewness

- ▶ SK is a **sample estimate** of $\bar{\mu}_3^s$.
- ▶ If sample skewness is zero, the distribution is symmetric.
- ▶ If SK is negative, more likely to get **large negative** returns than positive.
- ▶ If SK is positive, more likely to obtain **large positive** returns than negative.
- ▶ Most stock returns exhibit negative skewness.

Overall Likelihood of Extreme Returns?

- ▶ Need a measure of likely extreme events (positive or negative).
- ▶ Sample Kurtosis

$$KT = \frac{1}{T} \sum_{t=1}^T \left(\frac{r_t - \bar{r}}{s} \right)^4$$

- ▶ KT meaningless in itself. Need a benchmark. We usually use the normal and often consider excess kurtosis.

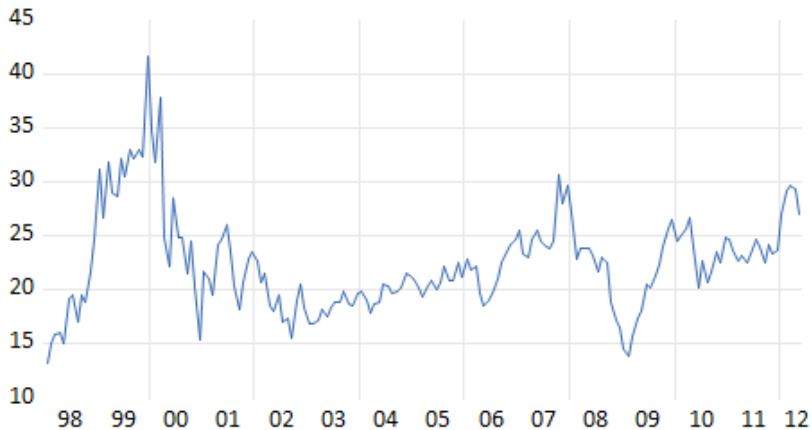
$$\text{Ex. Kurt} = KT - 3$$

- ▶ If $\text{Ex. Kurt} > 0$, tails are thicker than normal $\implies$ more extreme events than normal.
- ▶ If $\text{Ex. Kurt} < 0$, tails are thinner than normal $\implies$ fewer extreme events than normal.

Microsoft stock and the S&P 500 index

We illustrate the descriptive statistical analysis of financial data using monthly closing prices on Microsoft stock and the S&P 500 index over the period January 1998 and May 2012.

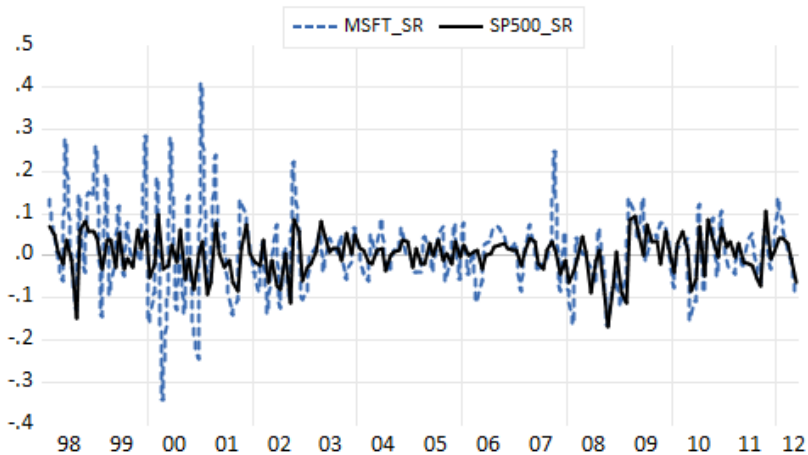
Prices on Microsoft stock, January 1998 to May 2012



S&P 500 index, January 1998 to May 2012



Monthly simple returns on Microsoft stock and the S&P 500 index.



Stationarity

The prices appear to be non-stationary since it has no tendency to revert to a time independent mean.

In contrast to prices, returns show clear mean-reverting behavior and the common monthly mean values look to be very close to zero.

Hence, the common mean value assumption of covariance stationarity looks to be satisfied.

The sample statistics for the Microsoft and S&P 500 monthly **simple** returns are given on the next slide.

Sample shape statistics

	MSFT	SP500
Mean	0.0092	0.0028
Variance	0.0103	0.0023
Std Dev	0.1015	0.0478
Skewness	0.4853	-0.5631
Ex. Kurt	1.9148	0.6545

The mean and standard deviation for Microsoft monthly returns are 0.9% and 10%, respectively.

Annualized, these values are approximately 10.8% ($.009 \times 12$) and 34.6% ($.10 \times \sqrt{12}$), respectively.

The corresponding monthly and annualized values for S&P 500 returns are 0.3% and 4.7%, and 4.8% and 16.6%, respectively.

Sample shape statistics

Microsoft has a higher mean and volatility than S&P 500.

The lower volatility for the S&P 500 reflects risk reduction due to diversification (holding many assets in a portfolio).

The sample skewness for Microsoft, 0.485, is slightly positive. The sample skewness for S&P 500, however, is moderately negative at -0.563, reflecting the somewhat long left tail of the histogram shown in Week 1.

The sample excess kurtosis values for Microsoft and S&P 500 are 1.915 and 0.655, respectively, and indicate that the tails are slightly fatter than the tails of a normal distribution.